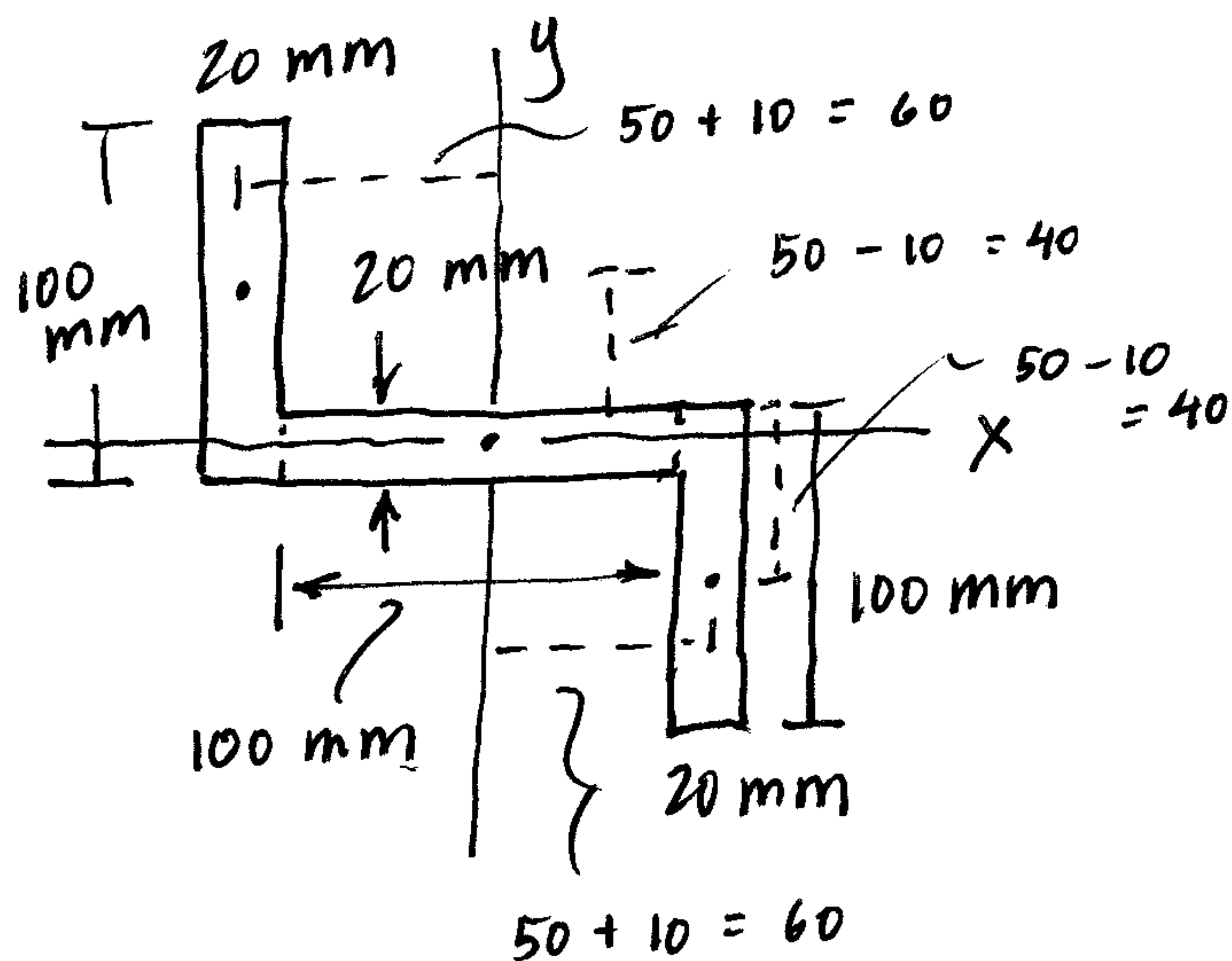


Composite Area MOI Example Problem 1



Determine I_x, I_y .

Use PAT:

$$I_{Axis} = \bar{I} + Ad^2$$

or

$$I_{Axis} = \sum (\bar{I} + Ad^2)$$

$$I_x = 2 \left[\frac{1}{12} (20)(100)^3 + (100)(20)(40)^2 \right] + \frac{1}{12} (100)(20)^3$$

$$I_x = 2 \left[1.67 \cdot 10^6 + 3.2 \cdot 10^6 \right] + 66.67 \cdot 10^3$$

$$I_x = 9.80 \cdot 10^6 \text{ mm}^4$$

$$I_y = 2 \left[\frac{1}{12} (100)(20)^3 + (100)(20)(60)^2 \right] + \frac{1}{12} (20)(100)^3$$

$$= 2 \left[66.67 \cdot 10^3 + 7.2 \cdot 10^6 \right] + 1.67 \cdot 10^6$$

$$I_y = 16.20 \cdot 10^6 \text{ mm}^4$$

$$I_{xy} = \sum (\bar{I}_{xy} + Adx dy)$$

Product of Inertia (POI)

$$= [0 + 100(20)(-60)(+40)] + 0 + [0 + 100(20)(+60)(-40)]$$

$$I_{xy} = -4.8 \cdot 10^6 \text{ mm}^4$$